Circulatory System

Blood

Circulatory System
- Blood
- Heart
- Blood vessels
- Lymph

1628
↳ William Harvey
↳ Double circulatory system in our Body

- Study of Blood → Angiology

Circulatory System

Open circulatory system

Arthropoda
↳ श्वेत → Blood
↳ organ
↓
organ.

mollusca → Snail,

Ex - insect → कॉकरोच, तितली

Closed circulatory system

Blood
↳ Blood vessels
↳ Heart → Blood pump

Human mammals

Blood

- Blood is a colloidal Sol. of 1nm - 100nm particle size.

Blood composition

- Water (90 - 92%)
- $O_2 + CO_2$
- vitamins
- Ca
- ~~cell~~ cells
- Protein
- Hormone
- Glucose
- Nutrient

Blood

- fluid.
- Blood is a connective Tissue
- Blood is not a organ it is a tissue
- Study of blood → Haematology

Blood

- Liquid (Plasma) 55%
- Solid (Corpuscells / cells) कोशिकाएं
  RBC, WBC, Platelets
  45%

Blood formation :- Bones (Adult)
(In child)
Blood formed in Liver, Spleen.

→ R.B.C.

(Red Bone marrow)

ADULT

Male	Female
(5-6) Litre	(4-5) Litre
(14g-16g Hb/100ml)	(12-13 g Hb/100ml)

PH → 7.4 (Basic)

Body weight (7%) → Blood.

colour :- RBC

Haemoglobin ⇒ बैंगनी
+ +
Iron Protien
+
Fe^{2+} +O_2 / O_2
Fe
O_2 O_2

• Hemoglobin contains Iron + Protien When this Fe comes contact with O_2 it terms red in colour.

• कॉकरोच → सफेद (Haemolymph)
• Octopus → नीला (Haemocyanin)
+
cobalt (नीला)

Humans

Oxygenated

Hb → Oxyhemoglobin

RBC → O_2

Pure blood

It flows in Arteries

Ex. P.A.

De-oxygenated

Hb → carboxy hemoglobin

RBC → CO_2

Impure Blood

Veins

$\boxed{\text{Ex. P.V}}$

• PLASMA

↳ Colour

↳ Pale yellow

↳ Billibrubin / Billivendin

↳ Pigment (Liver)

↳ 90 - 92 % Water

↳ 1 - 2 % ions e.g. Ca^{++} / mg^{++} / cl^{-}

↳ Glucose (Nutrient)

↳ Protien (8 - 9 %)

Protien :-

↳ Thrombin ← Blood clotting
↳ Prothrombin ← Blood clotting
↳ Fibrin ← Blood clotting
↳ fibrinogen ← Blood clotting
↳ Albumin → Salt / ions / mineral → Balancing
↳ Globulin → Antibodies / Immunity
↳ Hb → (O_2 Transport)

RBc

→ Red Blood corpuscles / cells

→ Erythro cyte
↓ ↓
Red cell

formation
↳ Erythropoiesis

→ A.V Leuwehadk
→ Embryo → Liver/spleen ; RBc → R.B.A

$\boxed{RBc > PL > WBC}$

Shape

→ Biconcave

Largest RBc → Amphuma
Smallest " → musk deer
mammals Largest → Elephant

RBC :—

Male (5- 5.5 million/100ml)

PL + WBC → (M+F) → same

Life span → 120 days → Liver / spleen

↓

(RBC का कब्रिस्तान)

लड़कियों (4 – 4.5 million/100 ml)

Natural Blood Bank → Spleen

↓

Blood ↓↓

↓

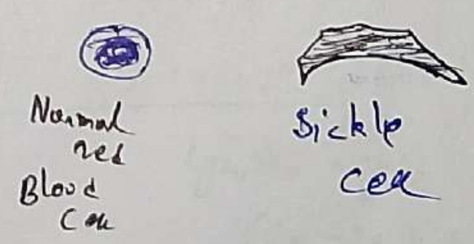
RBC बनाने मे भी मदद करता है

Erythroblastosis Foetalis

→ Same blood group

• Sickle cell Anemia

• It is genetic problem



• WBC → White Blood cells.

• other name → Leuco - cyto
↓ Leuco → White
↓ cyto → cell

• खोज :- Andral & Addison

• संख्या :- 8 - 10k /100ml of Hb.

• RBC : WBC ≃ 600 : 1

• Life - span = 12 - 20 days

Death :-
↳ Blood

WBC Nucleus :- (✓)

Shape → Irregular (Amoeba)

other name :-

① Fighter cell → phagocytic cell.
↳ phagocytosis

② Bodyguard cell → WBC
↳ Protection from infection

③ Soldier cell → WBC

Types of WBC

Granulocyte → cytoplasm → $\boxed{\text{Particle}}$ → Lobe

A-Granulocyte → No Particle

Granulocyte

Eosinophil 
(2-3%)
WBC
↓
Acidic medium
↓
Orange pink
$\boxed{\text{Bilobed}}$

Basophil 
Allergy
(0-1%)
WBC
↓
Basic medium
↓
Blue colour
$\boxed{\text{1, Bilobed}}$

Neutrophil
(60-65%) 
↓
Acid + Basic
↓
No colour
$\boxed{\text{3-5 lobe}}$

A-Granulocyte

Monocyte
(6-8%)

Lympho-cyte
(20-25%)